



Laxmi Charitable Trust's Sheth L.U.J College of Arts & Sir M.V. College Of Science & Commerce

(M2)Practical No. 14

Aim: Perform basic time series analysis in Scala. Generate synthetic time series data (e.g., daily sales over a month).

Code:

```
1 import scala.util.Random
2
3 object Shravan_M2_Prac14 {
4
5   def main(angs: Array[String]): Unit = {
6
7     val days = 1 to 30
8
9     val random = new Random(42)
10
11     val salesData = days.map { day =>
12       val sales = 5000 + (day * 100) + random.nextInt(1500)
13       (day, sales)
14     }
15
16     println("Daily Sales Time Series")
17     println("-----")
18
19     salesData.foreach {
20       case (day, sales) =>
21         println(f"Day $day%2d : ₹$sales%d")
22     }
23
24     val totalSales = salesData.map(_._2).sum
```

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Roll No: S117
Subject: Scala In DS



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```
object Shrvan_M2_Prac14 {  
  def main(args: Array[String]): Unit = {  
    val totalSales = salesData.map(_._2).sum  
  
    val averageSales = totalSales.toDouble / salesData.length  
  
    val minSales = salesData.minBy(_._2)  
  
    val maxSales = salesData.maxBy(_._2)  
  
    println("\nTime Series Analysis")  
    println("-----")  
    println(s"Total Sales : $totalSales")  
    println(s"Average Sales : $averageSales")  
    println(s"Minimum Sales : Day ${minSales._1} - ${minSales._2}")  
    println(s"Maximum Sales : Day ${maxSales._1} - ${maxSales._2}")  
  
    // Determine trend  
    val firstDaySales = salesData.head._2  
    val lastDaySales = salesData.last._2  
  
    if (lastDaySales > firstDaySales) {  
      println("Overall Trend : Increasing")  
    } else if (lastDaySales < firstDaySales) {  
      println("Overall Trend : Decreasing")  
    } else {  
      println("Overall Trend : Stable")  
    }  
  }  
}
```

```
object Shrvan_M2_Prac14 {  
  def main(args: Array[String]): Unit = {  
    println(s"Average Sales : $averageSales")  
    println(s"Minimum Sales : Day ${minSales._1} - ${minSales._2}")  
    println(s"Maximum Sales : Day ${maxSales._1} - ${maxSales._2}")  
  
    // Determine trend  
    val firstDaySales = salesData.head._2  
    val lastDaySales = salesData.last._2  
  
    if (lastDaySales > firstDaySales) {  
      println("Overall Trend : Increasing")  
    } else if (lastDaySales < firstDaySales) {  
      println("Overall Trend : Decreasing")  
    } else {  
      println("Overall Trend : Stable")  
    }  
  }  
}
```

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Output:

```
PS Practicals S117 Version Control Current File Shrvan_M2_Prac14.scala
Run TimeSeriesAnalysis
c:\Users\admin\jdk\temurin-17.0.20\bin\java.exe ...
Daily Sales Time Series
-----
Day 1 : ₹6230
Day 2 : ₹6463
Day 3 : ₹6548
Day 4 : ₹6284
Day 5 : ₹6970
Day 6 : ₹5625
Day 7 : ₹5705
Day 8 : ₹7218
Day 9 : ₹5919
Day 10 : ₹6593
Day 11 : ₹6282
Day 12 : ₹6202
Day 13 : ₹6576
Day 14 : ₹7192
Day 15 : ₹6976
Day 16 : ₹6632
Day 17 : ₹7156
Day 18 : ₹7970
Day 19 : ₹8143
```

```
PS Practicals S117 Version Control Current File Shrvan_M2_Prac14.scala
Run TimeSeriesAnalysis
Day 20 : ₹7209
Day 21 : ₹8500
Day 22 : ₹8563
Day 23 : ₹8526
Day 24 : ₹8813
Day 25 : ₹8243
Day 26 : ₹8341
Day 27 : ₹8730
Day 28 : ₹8758
Day 29 : ₹9287
Day 30 : ₹8246

Time Series Analysis
-----
Total Sales : ₹219900
Average Sales : ₹7330.00
Minimum Sales : Day 6 - ₹5625
Maximum Sales : Day 29 - ₹9287
Overall Trend : Increasing

Process finished with exit code 0
```

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(M2)Practical No. 15

Aim: Create polynomial features from a dataset. Given a list of numbers (e.g., [1, 2, 3]), generate polynomial features up to degree 3 (e.g., [1, 1², 1³, 2, 2², 2³, 3, 3², 3³]).

Code:

```
1 object Shravan_M2_Prac15 {
2
3   def main(args: Array[String]): Unit = {
4
5     val salesData = List(10, 15, 20, 25, 30, 35, 40, 45, 50, 55)
6
7     val degree = 3
8
9     val polynomialFeatures = salesData.flatMap { value =>
10       (1 to degree).map { power =>
11         Math.pow(value, power).toInt
12       }
13     }
14
15     println("Original Sales Dataset:")
16     println(salesData)
17
18     println("\nPolynomial Features up to Degree 3:")
19     println(polynomialFeatures)
20
21     println("\nDetailed Polynomial Features:")
22     println("Sales\tSales^2\tSales^3")
23     println("-----")
24 }
```

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```
1 object Shravan_M2_Prac15 {  
2   def main(args: Array[String]): Unit = {  
3     println("Original Sales Dataset:")  
4     println(salesData)  
5  
6     println("\nPolynomial Features up to Degree 3:")  
7     println(polynomialFeatures)  
8  
9     println("\nDetailed Polynomial Features:")  
10    println("Sales\tSales^2\tSales^3")  
11    println("-----")  
12  
13    salesData.foreach { value =>  
14      val square = value * value  
15      val cube = value * value * value  
16  
17      println(s"$value\t$square\t$cube")  
18    }  
19  }  
20}
```

Output:

```
Original Sales Dataset:  
List(10, 15, 20, 25, 30, 35, 40, 45, 50, 55)  
  
Polynomial Features up to Degree 3:  
List(10, 100, 1000, 15, 225, 3375, 20, 400, 8000, 25, 625, 15625, 30, 900, 27000, 35, 1225, 42875, 40, 1600, 64000, 45, 2025, 91125, 50, 2500, 125000, 55, 3025, 166375)  
  
Detailed Polynomial Features:  
Sales Sales^2 Sales^3  
-----  
10 100 1000  
15 225 3375  
20 400 8000  
25 625 15625  
30 900 27000  
35 1225 42875  
40 1600 64000  
45 2025 91125  
50 2500 125000  
55 3025 166375  
  
Process finished with exit code 0
```

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